BEE 4750 Homework 4: Linear Programming and Capacity Expansion

Name:

ID:

Due Date

Thursday, 11/06/25, 9:00pm

Overview

Instructions

- Problem 1 asks you to analytically (or graphically) solve a linear program.
- Problem 2 asks you to formulate and solve a resource allocation problem using linear programming.
- Problem 3 asks you to formulate, solve, and analyze a standard generating capacity expansion problem.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```
  Activating project at `c:\Users\sarah\OneDrive\Documents\GitHub\hw4-sarah-and-nathan-1`
```

```julia
using JuMP
using HiGHS
using DataFrames
using Plots
using Measures
using CSV
using MarkdownTables
```

Problems (Total: 30 Points)

Problem 1 (Total: 3 Points)

Solve the following linear program **analytically**. Show your work and justify any reasoning.

$$
\begin{aligned}
\max_{x,y} \quad & 3x + 2y \\
\text{subject to:} \quad & x + 4y \leq 16 \\
& x - 2y \leq -1 \\
& 0 \leq x \leq 4 \\
& 1 \leq y \leq 4.
\end{aligned}
$$

Problem 2 (12 points)

A farmer has access to a pesticide which can be used on corn, soybeans, and wheat fields, and costs $70/kg-yr to apply. The crop yields the farmer can obtain following crop yields by applying varying rates of pesticides to the field are shown in Table 1.

Application Rate (kg/ha)	Soybean (kg/ha)	Wheat (kg/ha)	Corn (kg/ha)
0	2900	3500	5900
1	3800	4100	6700
2	4400	4200	7900

Table 1: Crop yields from applying varying pesticide rates for Problem 1.

The costs of production, *excluding pesticides*, for each crop, and selling prices, are shown in Table 2.

Crop	Production Cost ($/ha-yr)	Selling Price ($/kg)
Soybeans	350	0.36
Wheat	280	0.27
Corn	390	0.22

Table 2: Costs of crop production, excluding pesticides, and selling prices for each crop.

Recently, environmental authorities have declared that farms cannot have an *average* application rate on soybeans, wheat, and corn which exceeds 0.8, 0.7, and 0.6 kg/ha, respectively. The farmer has asked you for advice on how they should plant crops and apply pesticides to maximize profits over 130 total ha while remaining in regulatory compliance if demand for each crop (which is the maximum the market would buy) this year is 250,000 kg?

Problem 2.1

Formulate a linear program for this resource allocation problem, including clear definitions of decision variable(s) (including units), objective function(s), and constraint(s) (make sure to explain functions and constraints with any needed derivations and explanations). **Tip: Make sure that all of your constraints are linear**.

Problem 2.2

How many ha should the farmer dedicate to each crop and with what pesticide application rate(s)? How much profit will the farmer expect to make?

Problem 2.3

The farmer has an opportunity to buy an extra 10 ha of land. How much extra profit would this land be worth to the farmer? Discuss why this value makes sense and under what conditions you would recommend the farmer should make the purchase.

Problem 2 (15 points)

For this problem, we will use hourly load (demand) data from 2013 in New York's Zone C (which includes Ithaca). The load data is loaded and plotted below in Figure 1.

```
#2.1
using JuMP
using HiGHS

#Data from question
farm_model = Model(HiGHS.Optimizer) #initializing farm model
crops = [:soy, :wheat, :corn] #initializing different crop types
Pest_rates = [0, 1, 2] #Application rates
Production_cost = Dict(:soy=>350, :wheat=>280, :corn=>390) #production cost of each crop $/ha
price = Dict(:soy=>0.36, :wheat=>0.27, :corn=>0.22) #Selling price $/kg
Pestrate_limit= Dict(:soy=>0.8, :wheat=>0.7, :corn=>0.6) #regulatory average limits (kg/ha)
pesticide_cost = 70 # $/kg
land_total = 130 #ha
demand = 250000 #kg/crop

#yields for each crop at each pesticide application rate (kg/ha)
yield_table = Dict(
  :soy => Dict(0=>2900, 1=>3800, 2=>4400),
  :wheat => Dict(0=>3500, 1=>4100, 2=>4200),
  :corn => Dict(0=>5900, 1=>6700, 2=>7900)
)

#area variables: A[i,r] = hectares of crop i planted at pesticide rate r
@variable(farm_model, A[i in crops, r in Pest_rates] >= 0)

#maximize profit = revenue - base costs - pesticide cost
@objective(farm_model, Max,
    sum( A[i,r] * ( price[i] * yield_table[i][r] - base_cost[i] - pesticide_cost * r )
         for i in crops, r in Pest_rates)
)

#land constraint
@constraint(farm_model, sum(A[i,r] for i in crops, r in Pest_rates) <= land_total)

#Demand cap per crop (total production of crop 1,2,3 <= demand)
@constraint(farm_model, [i in crops], sum(A[i,r] * yield_table[i][r] for r in Pest_rates) <=

#Regulatory application rate per crop:
@constraint(farm_model, [i in crops],
    sum(r * A[i,r] for r in Pest_rates) <= Pestrate_limit[i] * sum(A[i,r] for r in Pest_rates
)
```

```
#Solve
optimize!(farm_model)

#2.2

#Print the results of each crop
println("Objective (profit) = \$", objective_value(farm_model))
for i in crops
    println("Crop: ", i)
    for r in rates
        ha = value(A[i,r])
        if ha > 1
            println("rate = ", r, "kg/ha, area = ", ha, "ha, production = ", ha * yield_table
        end
    end
end

total_profit = objective_value(farm_model)
println("Total Profit: \$", total_profit)
```

```
UndefVarError: UndefVarError: `base_cost` not defined in `Main`
Suggestion: check for spelling errors or missing imports.
UndefVarError: `base_cost` not defined in `Main`

Suggestion: check for spelling errors or missing imports.



Stacktrace:

 [1] macro expansion

   @ C:\Users\sarah\.julia\packages\MutableArithmetics\VMS0x\src\rewrite_generic.jl:313 [inl

 [2] macro expansion

   @ C:\Users\sarah\.julia\packages\MutableArithmetics\VMS0x\src\rewrite.jl:374 [inlined]

 [3] macro expansion

   @ C:\Users\sarah\.julia\packages\JuMP\vfM5V\src\macros.jl:264 [inlined]
```

```
[4] macro expansion

  @ C:\Users\sarah\.julia\packages\JuMP\vfM5V\src\macros\@objective.jl:103 [inlined]

[5] macro expansion

  @ C:\Users\sarah\.julia\packages\JuMP\vfM5V\src\macros.jl:419 [inlined]

[6] top-level scope

  @ c:\Users\sarah\OneDrive\Documents\GitHub\hw4-sarah-and-nathan-1\jl_notebook_cell_df34fa9
```

```
#2.3
using JuMP
using HiGHS

#Data from question
farm_model = Model(HiGHS.Optimizer) #initializing farm model
crops = [:soy, :wheat, :corn] #initializing different crop types
Pest_rates = [0, 1, 2] #Application rates
Production_cost = Dict(:soy=>350, :wheat=>280, :corn=>390) #production cost of each crop $/ha
price = Dict(:soy=>0.36, :wheat=>0.27, :corn=>0.22) #Selling price $/kg
Pestrate_limit= Dict(:soy=>0.8, :wheat=>0.7, :corn=>0.6) #regulatory average limits (kg/ha)
pesticide_cost = 70 # $/kg
land_total = 140 #ha
demand = 250000 #kg/crop

#yields for each crop at each pesticide application rate (kg/ha)
yield_table = Dict(
  :soy => Dict(0=>2900, 1=>3800, 2=>4400),
  :wheat => Dict(0=>3500, 1=>4100, 2=>4200),
  :corn => Dict(0=>5900, 1=>6700, 2=>7900)
)

#area variables: A[i,r] = hectares of crop i planted at pesticide rate r
@variable(farm_model, A[i in crops, r in Pest_rates] >= 0)

#maximize profit = revenue - base costs - pesticide cost
@objective(farm_model, Max,
    sum( A[i,r] * ( price[i] * yield_table[i][r] - base_cost[i] - pesticide_cost * r )
         for i in crops, r in Pest_rates)
)
```

```
#land constraint
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# Regulatory application rate per crop:
@constraint(farm_model, [i in crops],
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)

#Solve
optimize!(farm_model)

#Print the results of each crop
println("Objective (profit) = \$", objective_value(farm_model))
for i in crops
    println("Crop: ", i)
    for r in rates
        ha = value(A[i,r])
        if ha > 1
            println("rate = ", r, "kg/ha, area = ", ha, "ha, production = ", ha * yield_table
        end
    end
end

total_profit = objective_value(farm_model)
println("Total Profit: \$", total_profit)
```

```
UndefVarError: UndefVarError: `base_cost` not defined in `Main`
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Stacktrace:

 [1] macro expansion
```

```
  @ C:\Users\sarah\.julia\packages\MutableArithmetics\VMS0x\src\rewrite_generic.jl:313 [inli
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 [6] top-level scope
  @ c:\Users\sarah\OneDrive\Documents\GitHub\hw4-sarah-and-nathan-1\jl_notebook_cell_df34fa
```
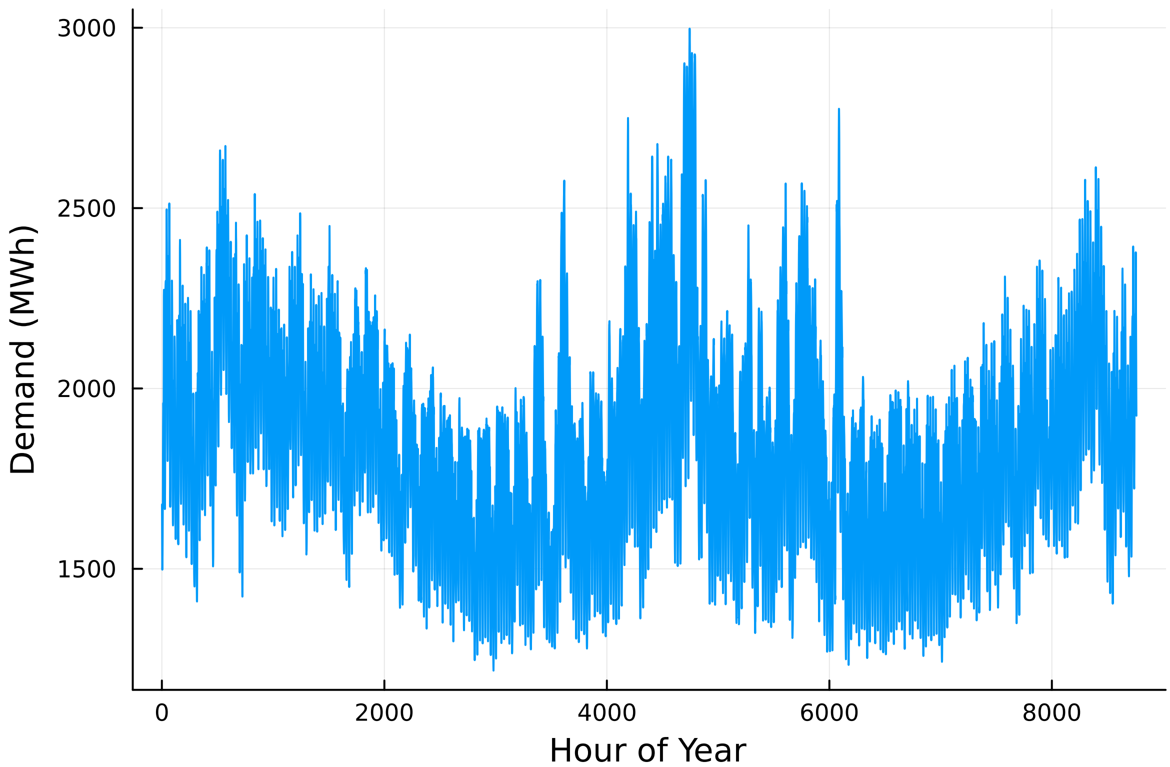

2.3) The farmer should expect to make around $7249 more dollars, adding another 10 ha of land. This increased profit comes from the ability to grow more wheat, as soy and corn are already maxed out in production. While we cannot sell any more corn or soy, we can sell more wheat. The extra 10ha does not maximize the wheat able to be sold, so this extra profit would make sense under certain conditions. These conditions include: if there is sufficient equipment to grow and harvest the crops, or if there is enough space in the trucks to move the wheat to a selling location.

```julia
# load the data, pull Zone C, and reformat the DataFrame
NY_demand = DataFrame(CSV.File("data/2013_hourly_load_NY.csv"))
rename!(NY_demand, :"Time Stamp" => :Date)
demand = NY_demand[:, [:Date, :C]]
rename!(demand, :C => :Demand)
demand[:, :Hour] = 1:nrow(demand)

# plot demand
plot(demand.Hour, demand.Demand, xlabel="Hour of Year", ylabel="Demand (MWh)", label=:false)
```



Next, we load the generator data, shown in Table 3. This data includes fixed costs ($/MW installed), variable costs ($/MWh generated), and CO2 emissions intensity (tCO2/MWh generated).

```
gens = DataFrame(CSV.File("data/generators.csv"))
```

	Plant	FixedCost	VarCost	Emissions
	String15	Int64	Int64	Float64
1	Geothermal	450000	0	0.0
2	Coal	220000	24	1.0
3	NG CCGT	82000	30	0.43
4	NG CT	65000	40	0.55
5	Wind	91000	0	0.0
6	Solar	70000	0	0.0

Finally, we load the hourly solar and wind capacity factors, which are plotted in Figure 2. These tell us the fraction of installed capacity which is expected to be available in a given hour for generation (typically based on the average meteorology).

```julia
# load capacify factors into a DataFrame
cap_factor = DataFrame(CSV.File("data/wind_solar_capacity_factors.csv"))

# plot January capacity factors
p1 = plot(cap_factor.Wind[1:(24*31)], label="Wind")
plot!(cap_factor.Solar[1:(24*31)], label="Solar")
xaxis!("Hour of the Month")
yaxis!("Capacity Factor")

p2 = plot(cap_factor.Wind[4344:4344+(24*31)], label="Wind")
plot!(cap_factor.Solar[4344:4344+(24*31)], label="Solar")
xaxis!("Hour of the Month")
yaxis!("Capacity Factor")

display(p1)
display(p2)
```

You have been asked to develop a generating capacity expansion plan for the utility in Riley County, NY, which currently has no existing electrical generation infrastructure. The utility can build any of the following plant types: geothermal, coal, natural gas combined cycle gas turbine (CCGT), natural gas combustion turbine (CT), solar, and wind. The NY state legislature is planning to enact an annual CO2 limit, which for the utility would limit the emissions in its footprint to 1.5 MtCO2/yr.

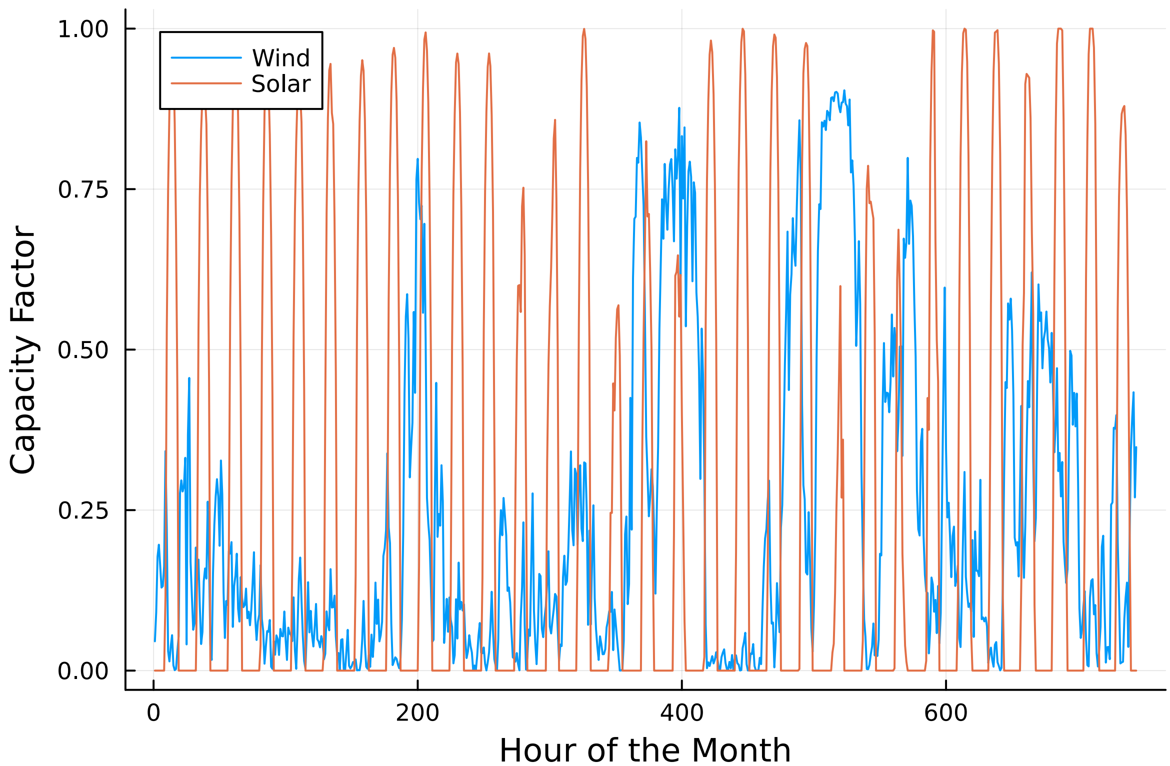
While coal, CCGT, and CT plants can generate at their full installed capacity, geothermal plants operate at maximum 85% capacity, and solar and wind available capacities vary by the hour depend on the expected meteorology. The utility will also penalize any non-served demand at a rate of $10,000/MWh.

Problem 3.1

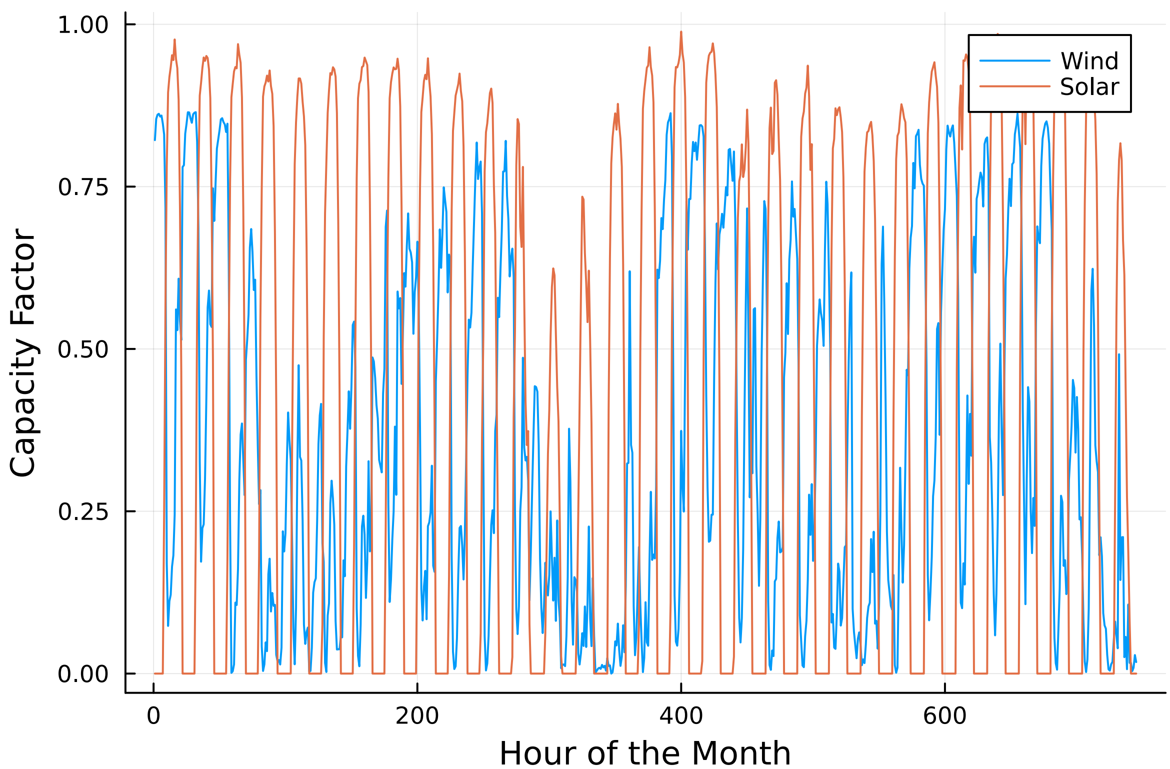
Formulate a linear program for this capacity expansion problem, including clear definitions of decision variable(s) (including units), objective function(s), and constraint(s) (make sure to explain functions and constraints with any needed derivations and explanations).

Problem 3.2

How much should the utility build of each type of generating plant? What will the total cost be? How much energy will be non-served?



Wind
Solar
Capacity Factor
1.00
0.75
0.50
0.25
0.00
0
200
400
600
Hour of the Month



Wind
Solar
Capacity Factor
1.00
0.75
0.50
0.25
0.00
0
200
400
600
Hour of the Month

Problem 3.3

What fraction of annual generation does each plant type produce? How does this compare to the breakdown of built capacity that you found in Problem 3.2? Do these results make sense given the demand and generator data?

Problem 3.4

Make a plot of the electricity price in each hour. Discuss any trends that you see.

Problem 3.5

What is the shadow price of CO2? What is the value to the utility be of allowing it to emit an additional 1000 tCO2/yr?

> **Significant Digits**
>
> Use `round(x; digits=n)` to report values to the appropriate precision! If your number is on a different order of magnitude and you want to round to a certain number of significant digits, you can use `round(x; sigdigits=n)`.
>
> **Getting Variable Output Values**
>
> `value.(x)` will report the values of a `JuMP` variable `x`, but it will return a special container which holds other information about `x` that is useful for `JuMP`. This means that you can't use this output directly for further calculations. To just extract the values, use `value.(x).data`.
>
> **Suppressing Model Command Output**
>
> The output of specifying model components (variable or constraints) can be quite large for this problem because of the number of time periods. If you end a cell with an `@variable` or `@constraint` command, I *highly* recommend suppressing output by adding a semi-colon after the last command, or you might find that your notebook crashes.

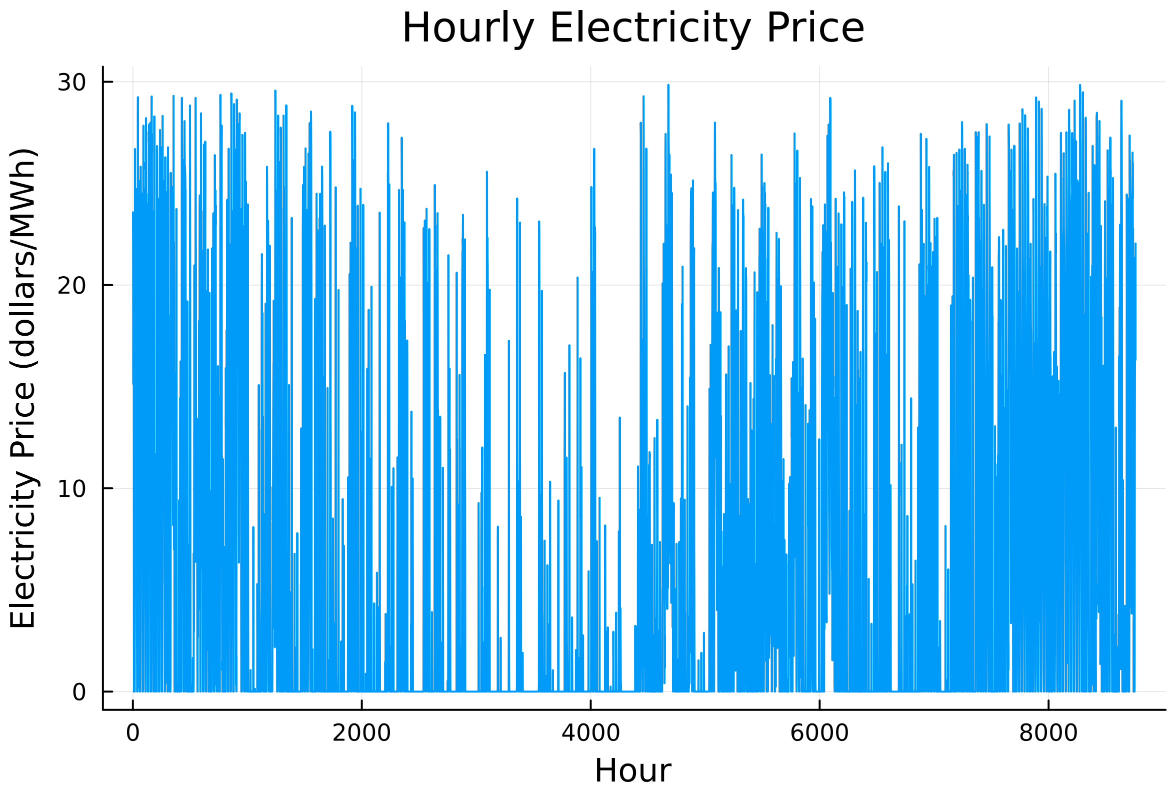
```julia
# 3.1 Solution

# Load data
T = 1:nrow(demand)  # hours in the year
G = gens.Plant         # generator names

fixed_cost = Dict(gens.Plant[i] => gens.FixedCost[i] for i in 1:nrow(gens))
var_cost = Dict(gens.Plant[i] => gens.VarCost[i] for i in 1:nrow(gens))
emissions = Dict(gens.Plant[i] => gens.Emissions[i] for i in 1:nrow(gens))
```

```julia
# Capacity factors for wind and solar defined from given data
cf = Dict((g, t) => (g == "Wind" ? cap_factor.Wind[t] :
                    g == "Solar" ? cap_factor.Solar[t] :
                    g == "Geothermal" ? 0.85 : 1.0) for g in G, t in T)

# Variables
cap_exp = Model(HiGHS.Optimizer)

@variable(cap_exp, x[g in G] >= 0)            # Built capacity for each generator (MW)
@variable(cap_exp, y[g in G, t in T] >= 0)    # Energy generation for each generator (MWh)
@variable(cap_exp, NSE[t in T] >= 0)          # Non-served energy (MWh)

# Objective

# Sum fixed cost for each generator multipled by capacity built, variable costs multipled by
# Results is total cost which must be minimized
@objective(cap_exp, Min,
    sum(fixed_cost[g] * x[g] for g in G) +
    sum(var_cost[g] * y[g, t] for g in G, t in T) +
    sum(10000 * NSE[t] for t in T)
)

# Constraints
@constraint(cap_exp, dem[t in T], sum(y[g, t] for g in G) + NSE[t] == demand[!, :Demand][t])
@constraint(cap_exp, cap[g in G, t in T], y[g, t] <= x[g] * cf[(g, t)])
@constraint(cap_exp, CO2, sum(emissions[g] * y[g, t] for g in G, t in T) <= 1.5e6)

optimize!(cap_exp)

# 3.2 solution

# Print and label results
println("\nBuilt Capacity (MW):")
for g in G
    println("$g: ", round(value(x[g]); digits=2))
end

println("\nTotal Annual Cost (dollars): ", round(objective_value(cap_exp); digits=2))

total_NSE = sum(value(NSE[t]) for t in T)
println("\nTotal Non-Served Energy (MWh): ", round(total_NSE; digits=2))
```

```julia
# 3.3 solution
total_gen = sum(value(y[g,t]) for t in T, g in G)

println("\nFraction of Annual Generation by Generator Type:")
for g in G
    println("$g: ", round(sum(value(y[g,t]) for t in T) / total_gen, digits=2))
end

println("\n3.3 explanation included in markdown cell below.")

# 3.4 solution

hourly_price = [sum(var_cost[g] * value(y[g, t]) for g in G) /
max(sum(value(y[g, t]) for g in G), 1e-6) for t in T]

# 3.5 solution
println("\nCO2 shadow price:", round(shadow_price(CO2); digits=2))

# Plot electricity price by hour
plot(T, hourly_price,
    xlabel = "Hour",
    ylabel = "Electricity Price (dollars/MWh)",
    title = "Hourly Electricity Price",
    legend = false)
```

```
Running HiGHS 1.11.0 (git hash: 364c83a51e): Copyright (c) 2025 HiGHS under MIT licence terms
LP   has 61321 rows; 61326 cols; 188256 nonzeros
Coefficient ranges:
  Matrix [5e-05, 1e+00]
  Cost   [2e+01, 4e+05]
  Bound  [0e+00, 0e+00]
  RHS    [1e+03, 2e+06]
Presolving model
56857 rows, 56862 cols, 179328 nonzeros  0s
Dependent equations search running on 8760 equations with time limit of 1000.00s
Dependent equations search removed 0 rows and 0 nonzeros in 0.00s (limit = 1000.00s)
56857 rows, 56862 cols, 179328 nonzeros  0s
Presolve : Reductions: rows 56857(-4464); columns 56862(-4464); elements 179328(-8928)
Solving the presolved LP
Using EKK dual simplex solver - serial
  Iteration        Objective     Infeasibilities num(sum)
          0     0.0000000000e+00 Pr: 8760(5.18855e+06) 0s
```

```
      21893     5.4327281307e+08 Pr: 9813(2.11986e+06); Du: 0(1.88022e-07) 5s
      27281     7.2226250425e+08 Pr: 13958(1.58071e+08); Du: 0(7.07015e-08) 10s
      30714     7.7989326602e+08 Pr: 2747(379512); Du: 0(7.07015e-08) 16s
      32698     7.8535295894e+08 Pr: 0(0); Du: 0(2.91607e-11) 19s
Solving the original LP from the solution after postsolve
Model status        : Optimal
Simplex   iterations: 32698
Objective value     :  7.8535295894e+08
P-D objective error :  1.3357583179e-14
HiGHS run time      :         18.77

Built Capacity (MW):
Geothermal: 285.57
Coal: 0.0
NG CCGT: 1509.17
NG CT: 703.06
Wind: 2548.89
Solar: 2103.75

Total Annual Cost (dollars): 7.8535295894e8

Total Non-Served Energy (MWh): 332.29

Fraction of Annual Generation by Generator Type:
Geothermal: 0.07
Coal: 0.0
NG CCGT: 0.2
NG CT: 0.01
Wind: 0.39
Solar: 0.33

3.3 explanation included in markdown cell below.

CO2 shadow price:-182.77
```



3.1 Comments throughout my model explain constraints, objective, and variables.

3.3 Explanation

This answer is consistent with the results I found for built capacity in 2.2. Wind has the largest built capacity in MW and largest generation in MWh. In contrast, coal has 0 MW of built capacity and thus 0 MWh of generation. Results are proportional for each generator in between. These results make sense given demand, cost, and the constraint on CO2 emissions. NG CCGT and NG CT have the highest variable costs and are constrained by their CO2 emissions. Geothermal has the highest fixed cost, limiting its built capacity. Coal has a high fixed cost and the highest emissions, so it is not utilized by this model at all. Wind and solar are the cheapest to build and have no variable costs or emissions production, so they are utilized most heavily. Other generators are used to meet demand when wind and solar have a low capacity factor.

3.4 Explanation

The data appears to cave inward, with lowest electricity prices found at 4000 hours. This may be due to higher utilization of wind and solar in these time periods.

3.5 Explanation

Allowing this model to emit an additional 1000 tCO2/yr reduces the overall cost by $182.77, as indicated by the negative shadow price.

References

List any external references consulted, including classmates.

Stack overflow for dict help Copilot used to fix syntax errors